

Time Series Analysis of Amazon Stock Price using MATLAB

Amazon.com is a multinational technology and e-commerce company based in Seattle, Washington. It was founded by Jeff Bezos in 1994 and is one of the largest companies in the world. Its operations include online retail, cloud computing, artificial intelligence, and logistics. Its stock is listed on the NASDAQ under the symbol AMZN.

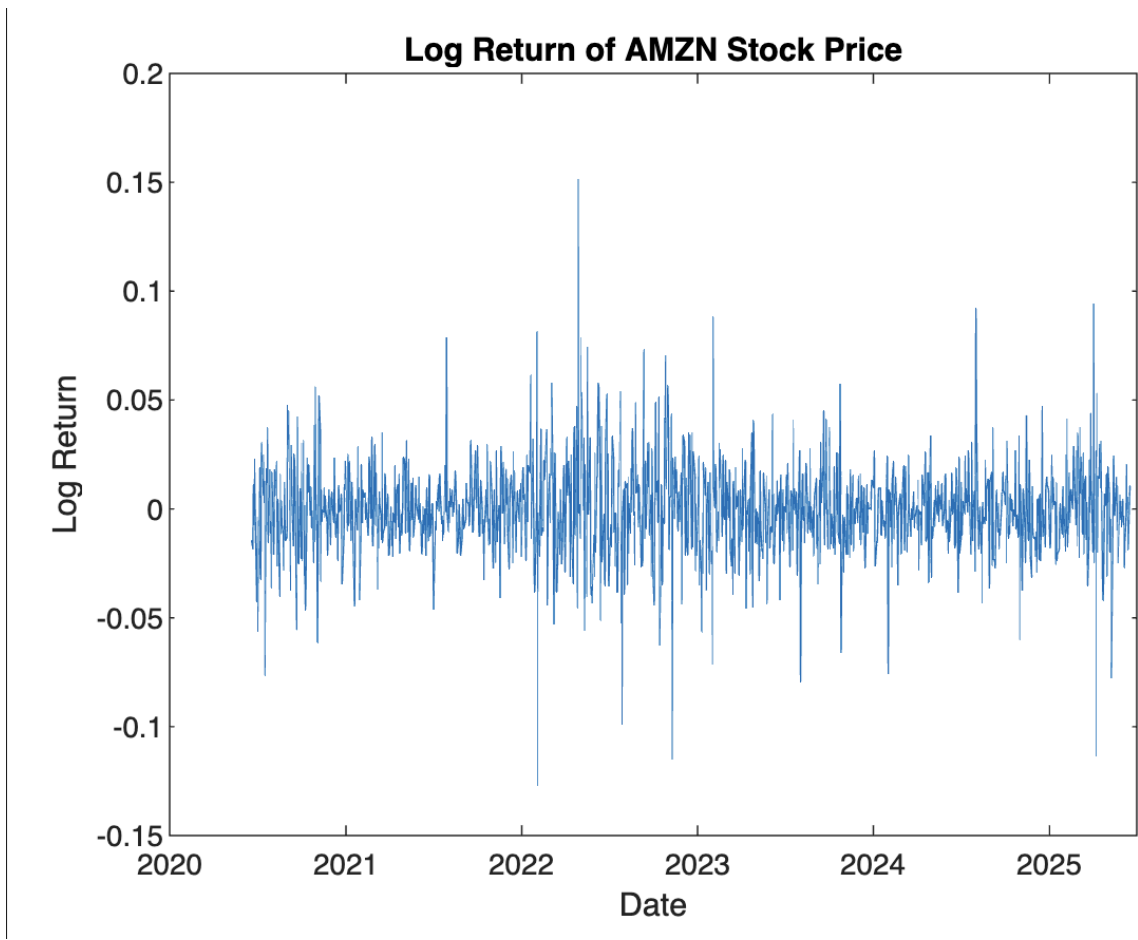
I picked Amazon as it is popular and widely followed stock. My goal is to apply time series techniques to understand Amazon stock price movements and effectiveness of various forecasting models. I used data for the last five years from April 19, 2020, to April 19, 2025 using Yahoo Finance.

I used a range of time series techniques, including ARMA, ARIMA and GARCH models, to understand Amazon stock returns over the five year period. Model performance is evaluated based on the model fit.

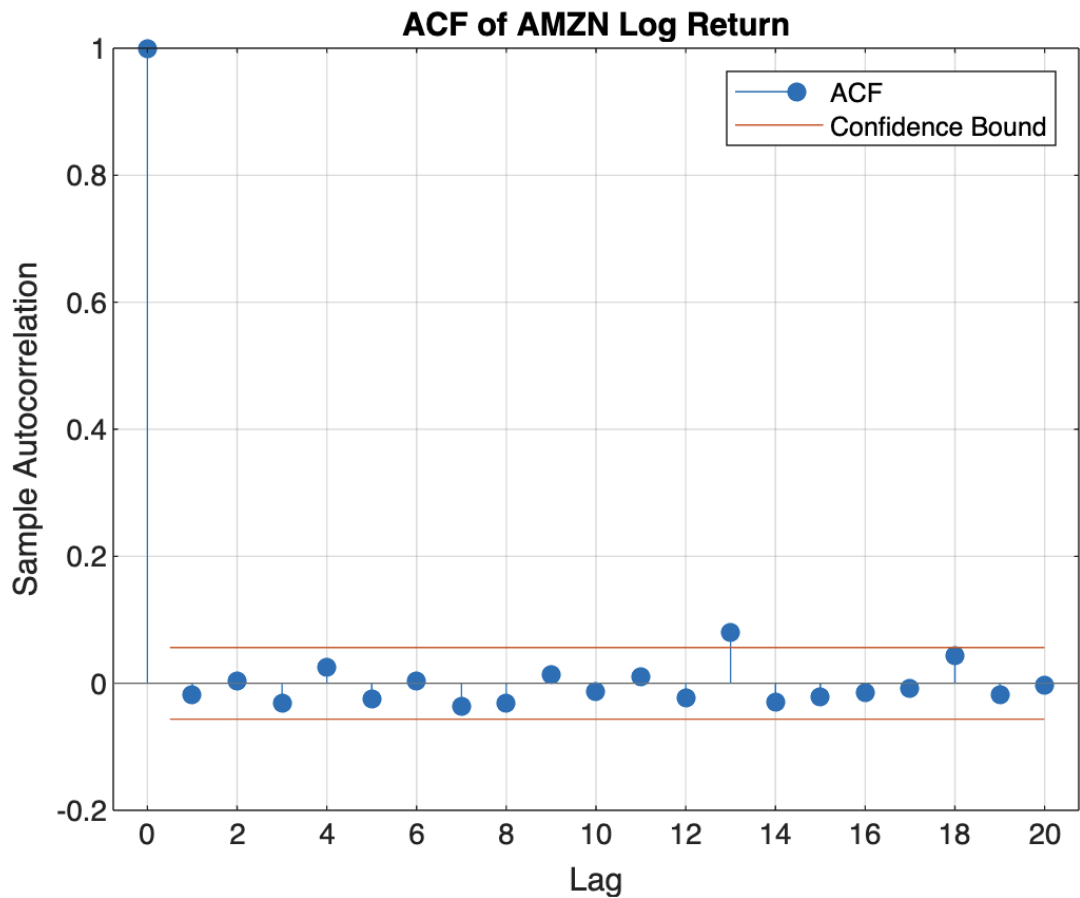
Time Series:



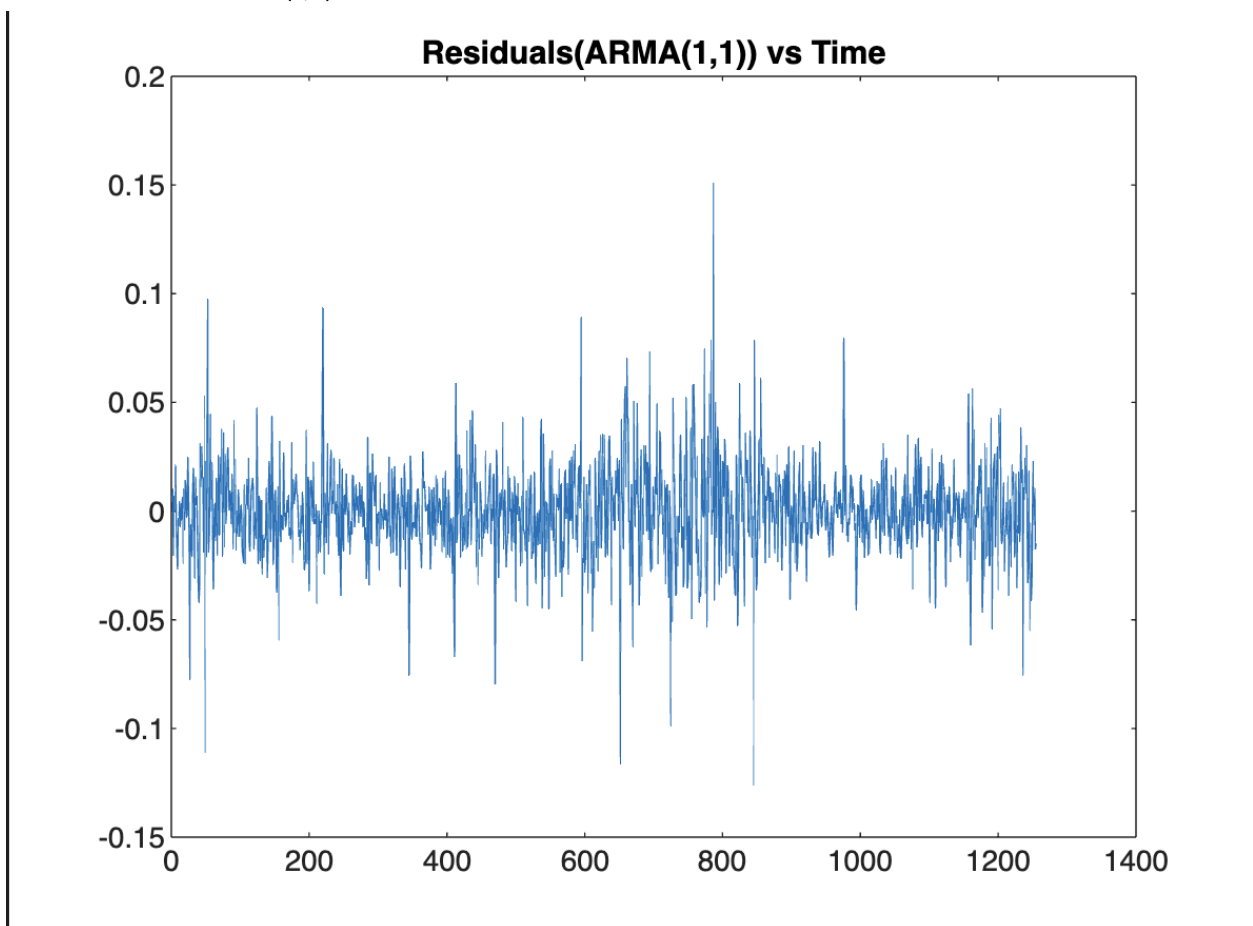
Log Return Graph



ACF of AMZN Log Returns:



Residuals of ARMA(1,1) Model:

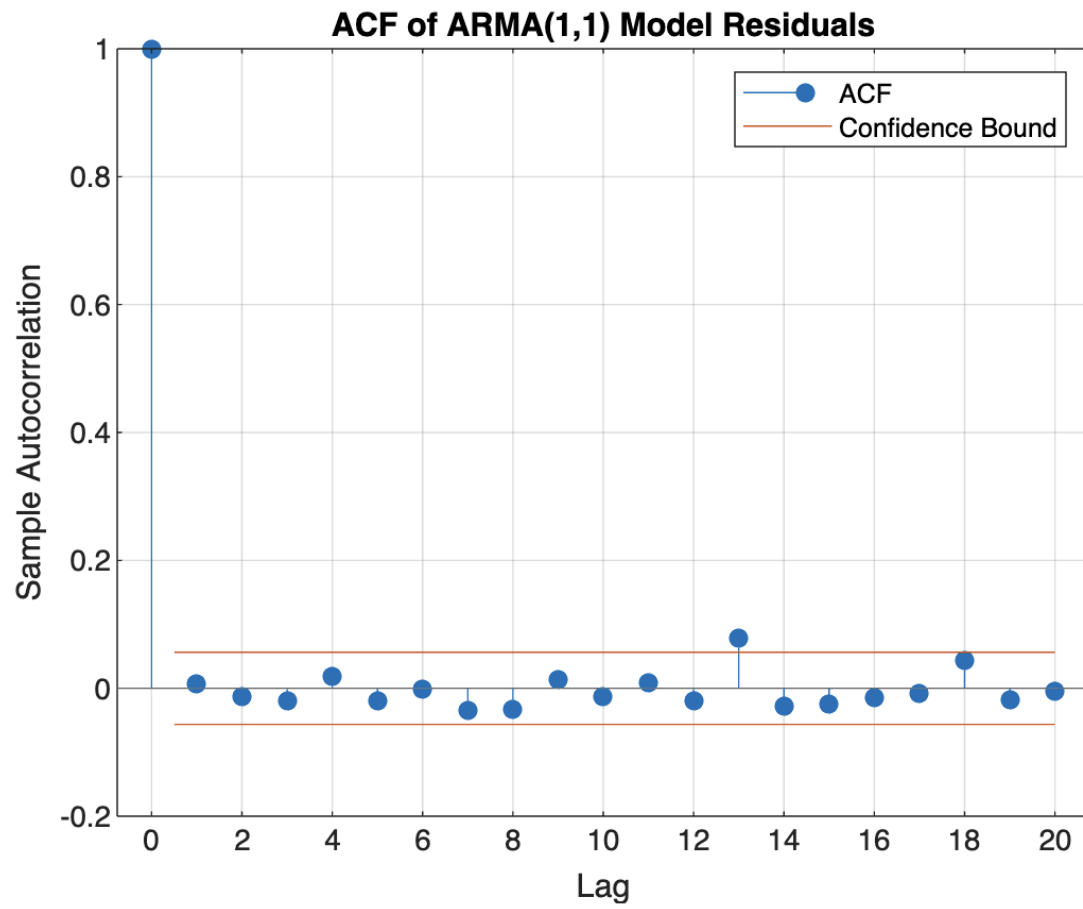


Results:

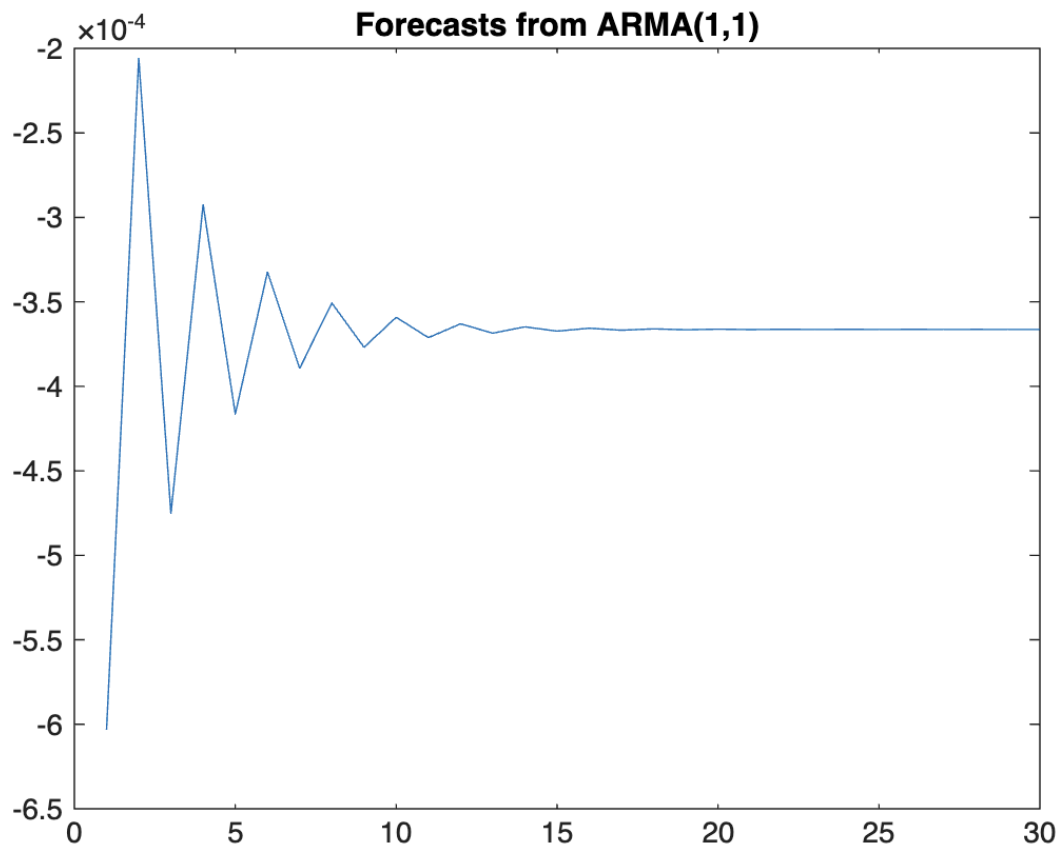
ARIMA(1,0,1) Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.00061468	0.0010531	-0.5837	0.55942
AR{1}	-0.67805	0.37628	-1.802	0.07155
MA{1}	0.65356	0.38802	1.6843	0.092117
Variance	0.00050471	1.1263e-05	44.811	0

ACF of ARMA(1,1) Residuals:



30-Day Forecast Using ARMA(1,1):

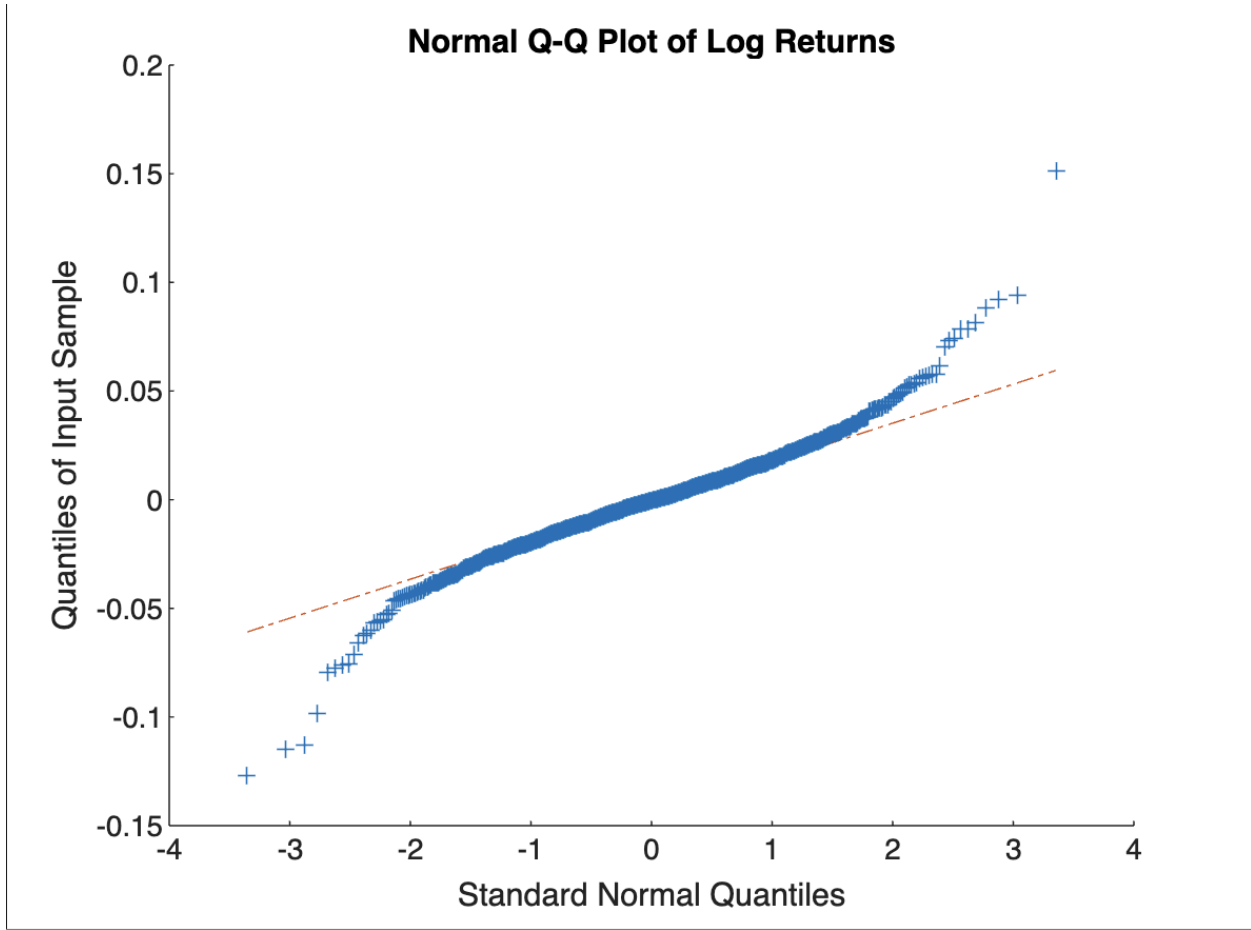


GARCH Model:

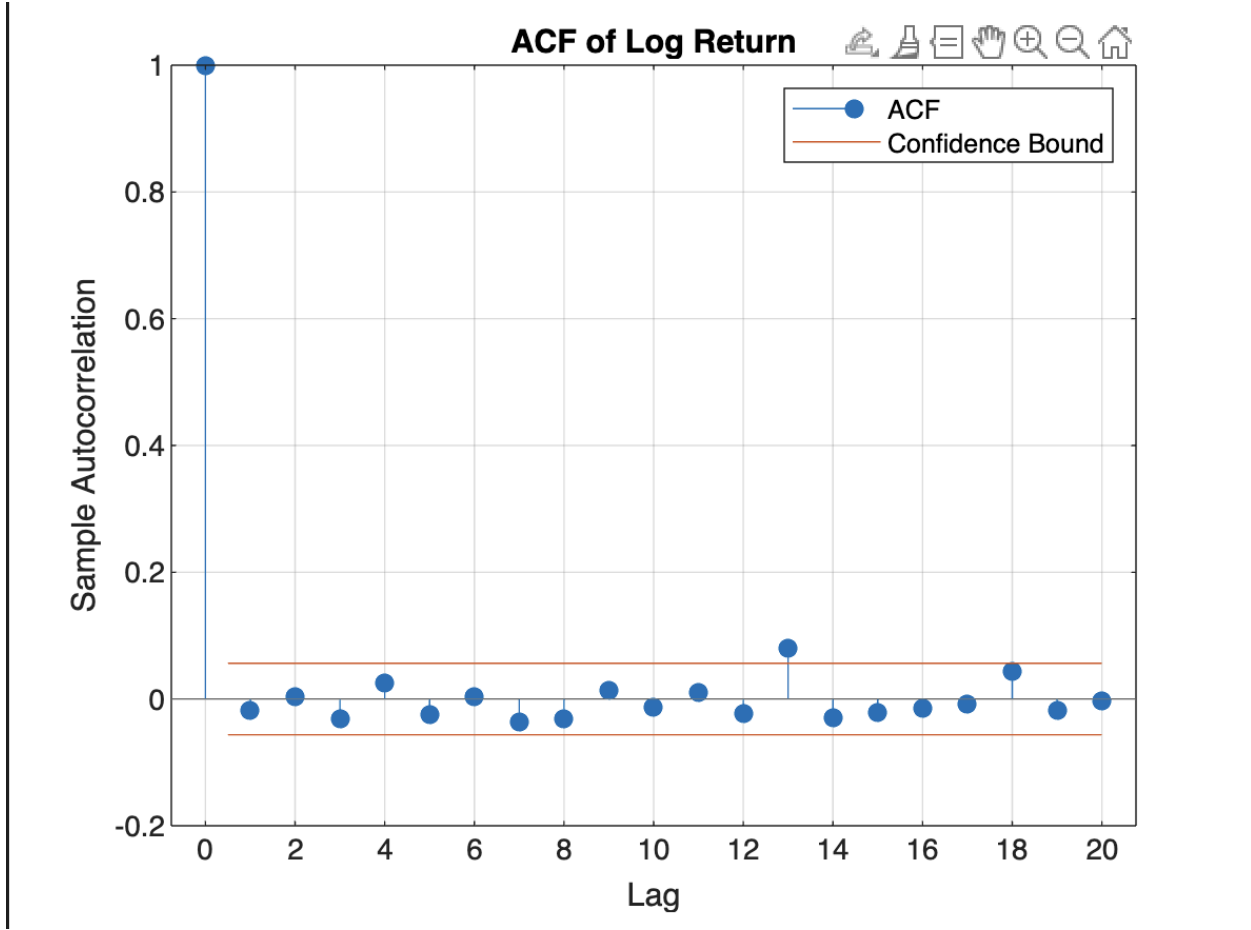
GARCH(1,1) Conditional Variance Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	1.8116e-05	3.2151e-06	5.6346	1.7543e-08
GARCH{1}	0.90034	0.014208	63.367	0
ARCH{1}	0.065379	0.010038	6.5131	7.3614e-11

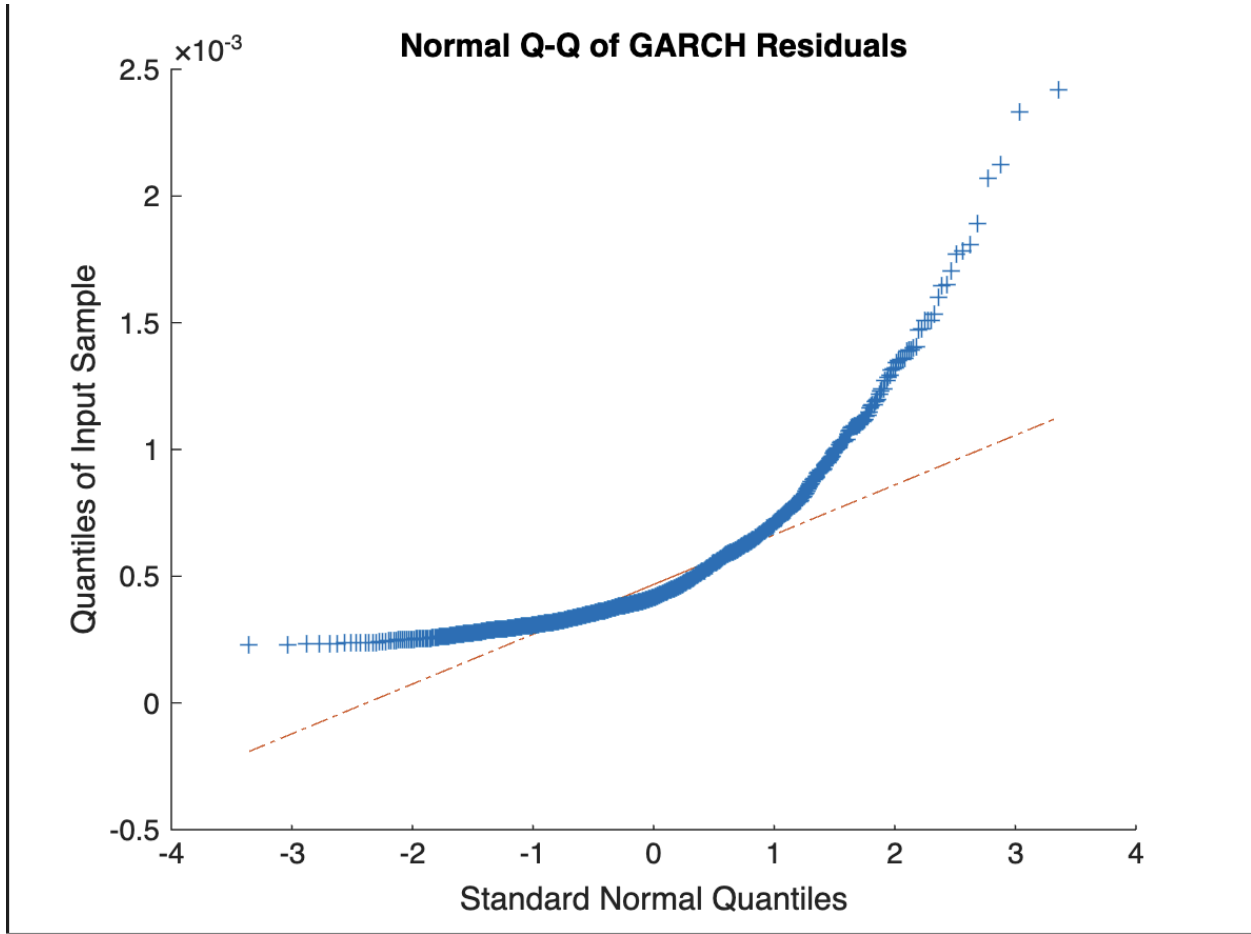
Normal Q-Q Plot of Log Returns:



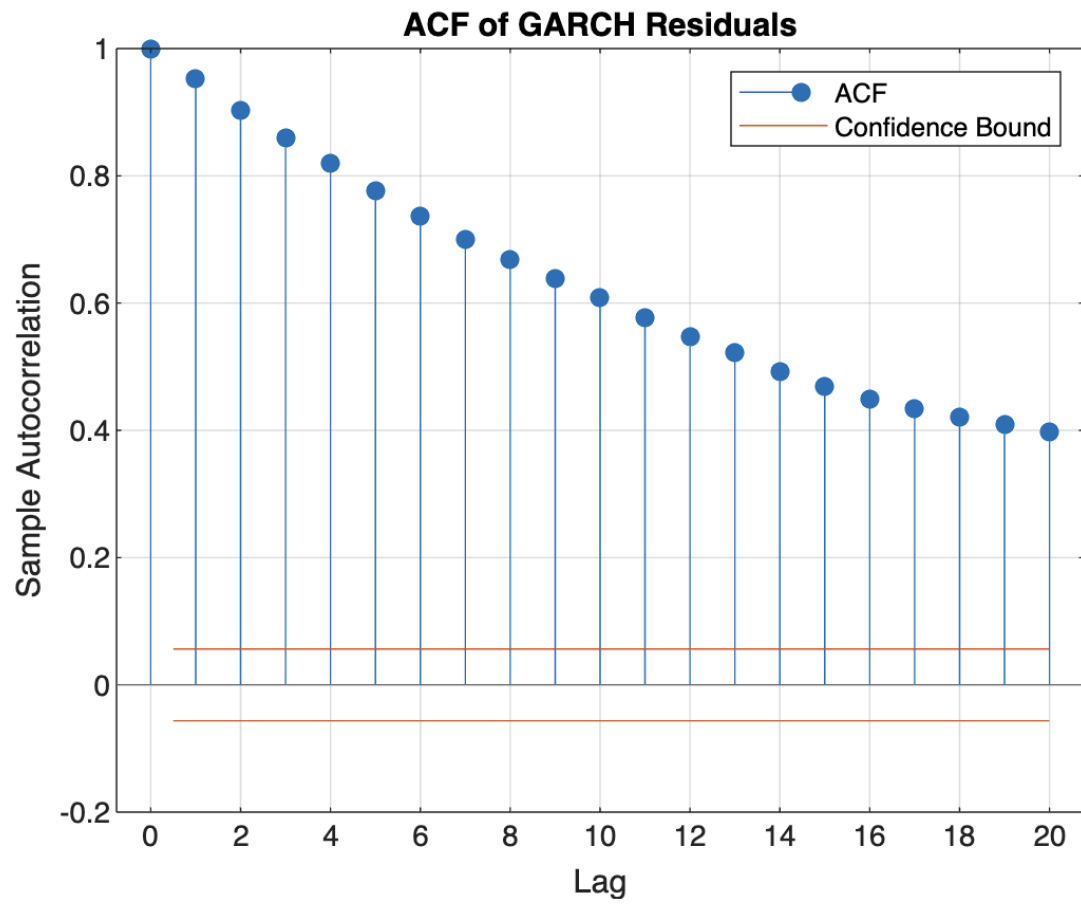
ACF of Log Returns:



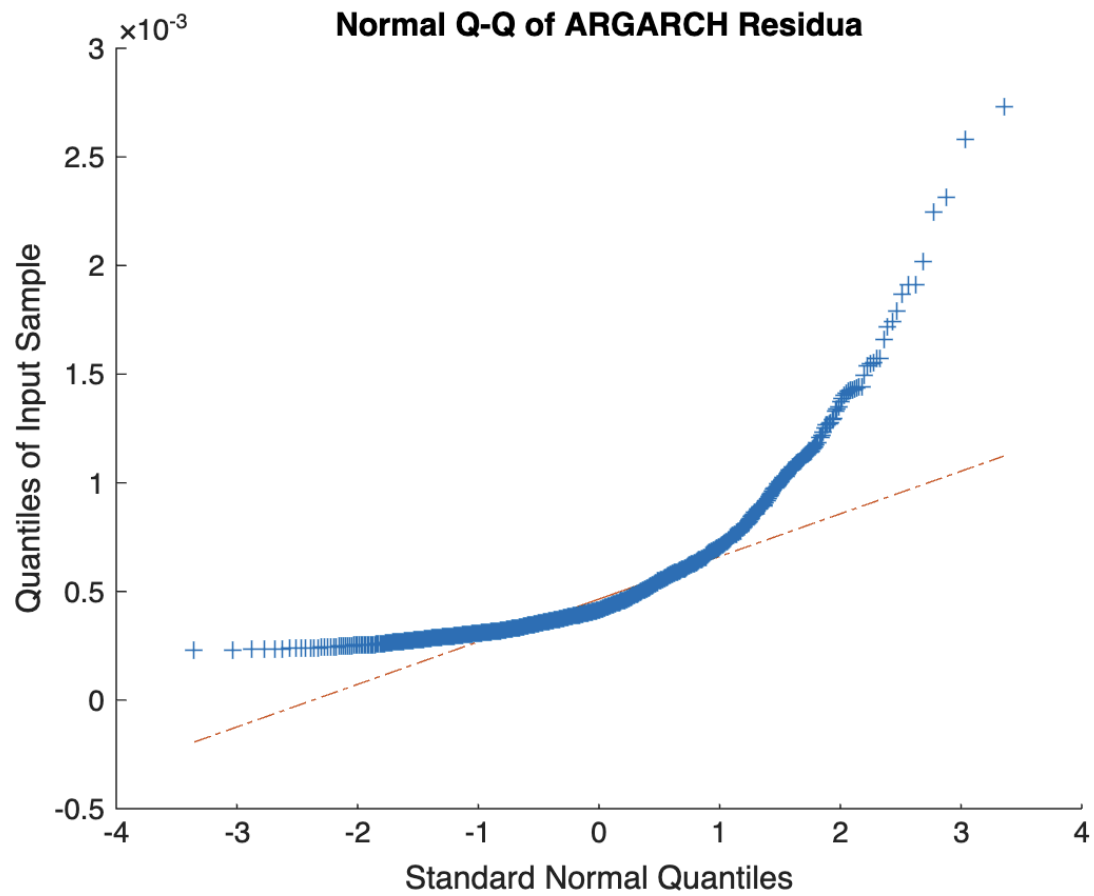
Normal Q-Q of GARCH Residuals:



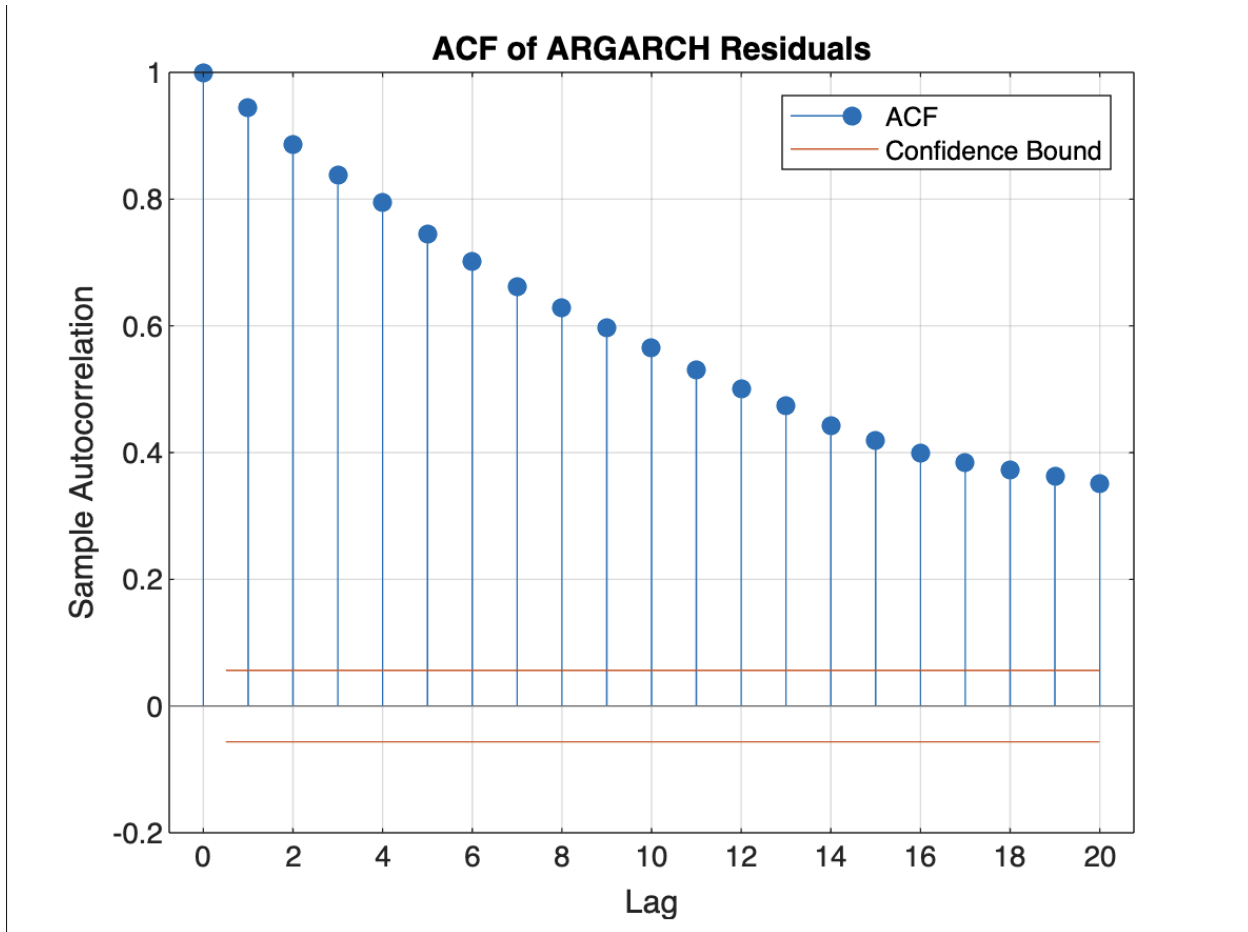
ACF of GARCH Residuals:



Asymmetric GARCH (e.g., GJR-GARCH):



ACF of ARGARCH Residuals:



GJR(1,1) Conditional Variance Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	2.1494e-05	3.9236e-06	5.4782	4.2957e-08
GARCH{1}	0.88608	0.016423	53.955	0
ARCH{1}	0.078535	0.015754	4.985	6.1966e-07
Leverage{1}	-0.0097798	0.019265	-0.50764	0.6117

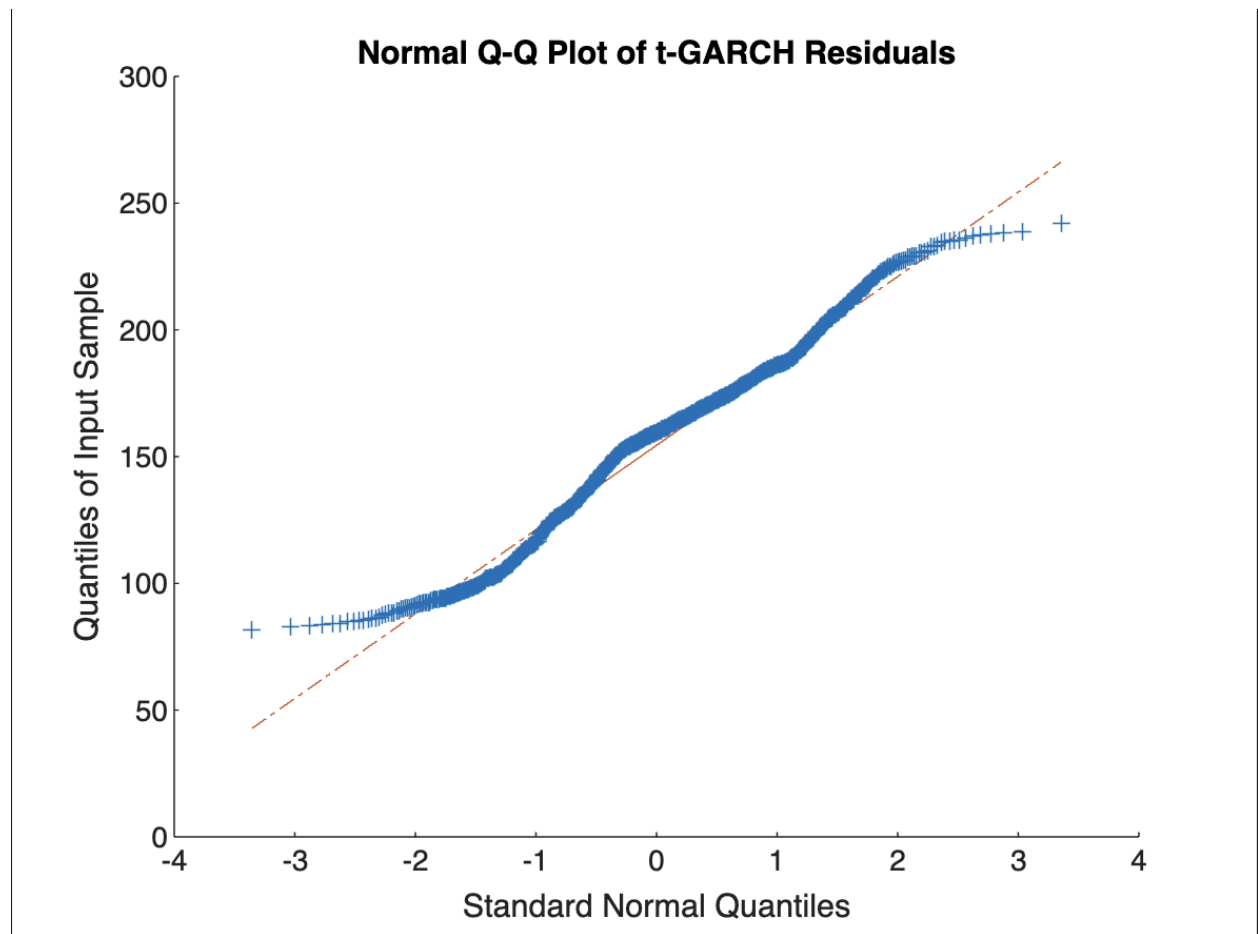
GARCH with t-distribution:

GARCH(1,1) Conditional Variance Model (t Distribution):

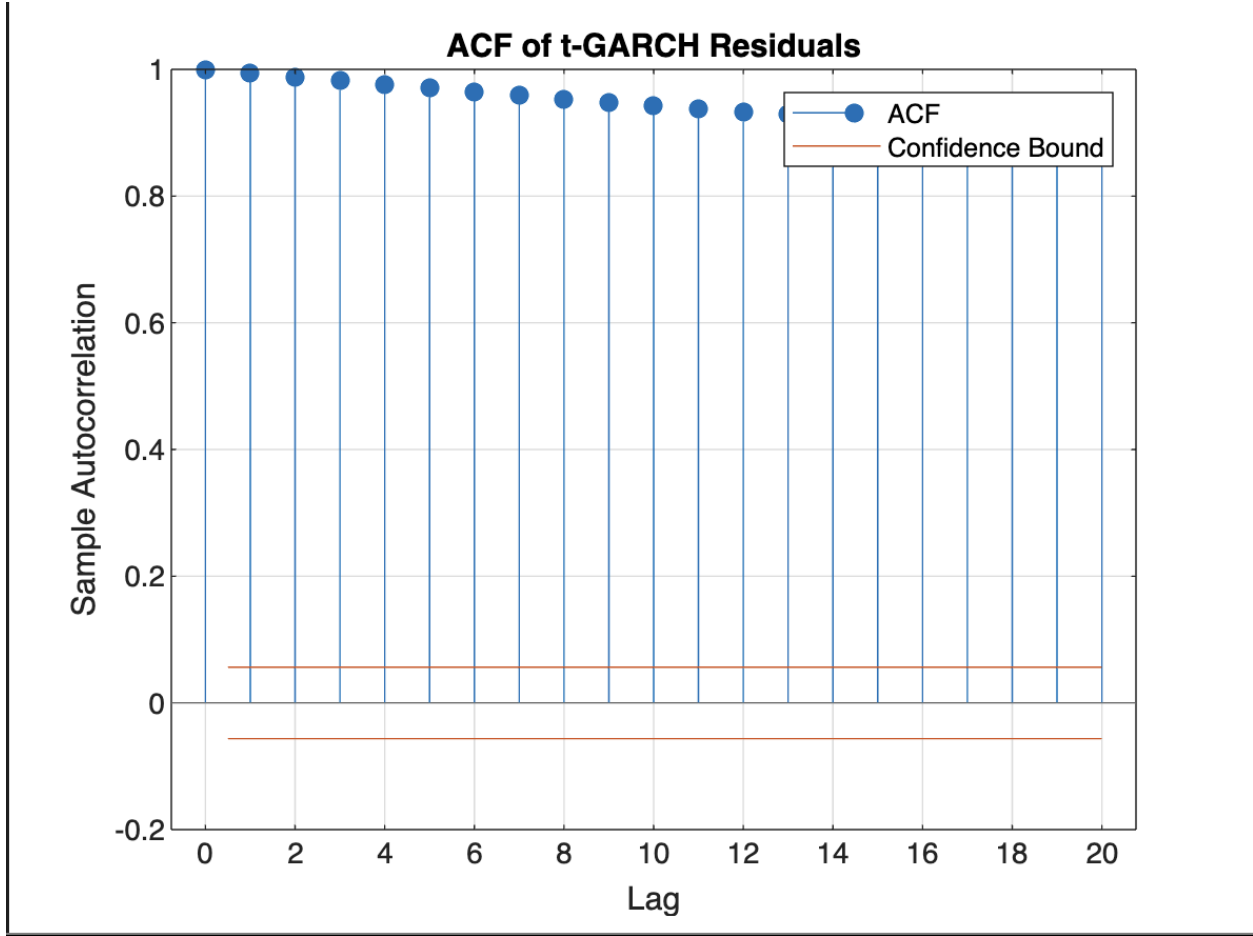
	Value	StandardError	TStatistic	PValue
Constant	9.2084e-06	4.0764e-06	2.2589	0.023887

GARCH{1}	0.92595	0.018924	48.929	0
ARCH{1}	0.058088	0.015275	3.8028	0.0001431
DoF	4.9725	0.62842	7.9127	2.5181e-15

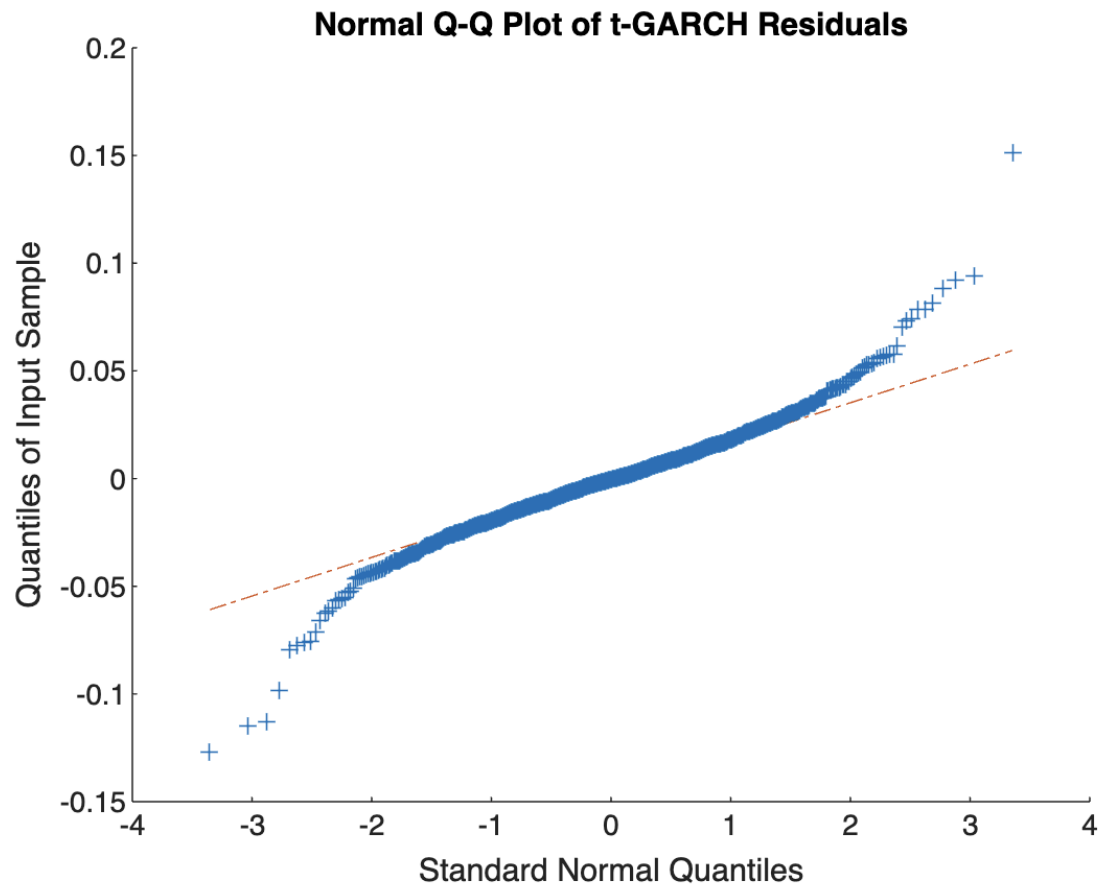
Normal Q-Q Plot of T-GARCH Residual:



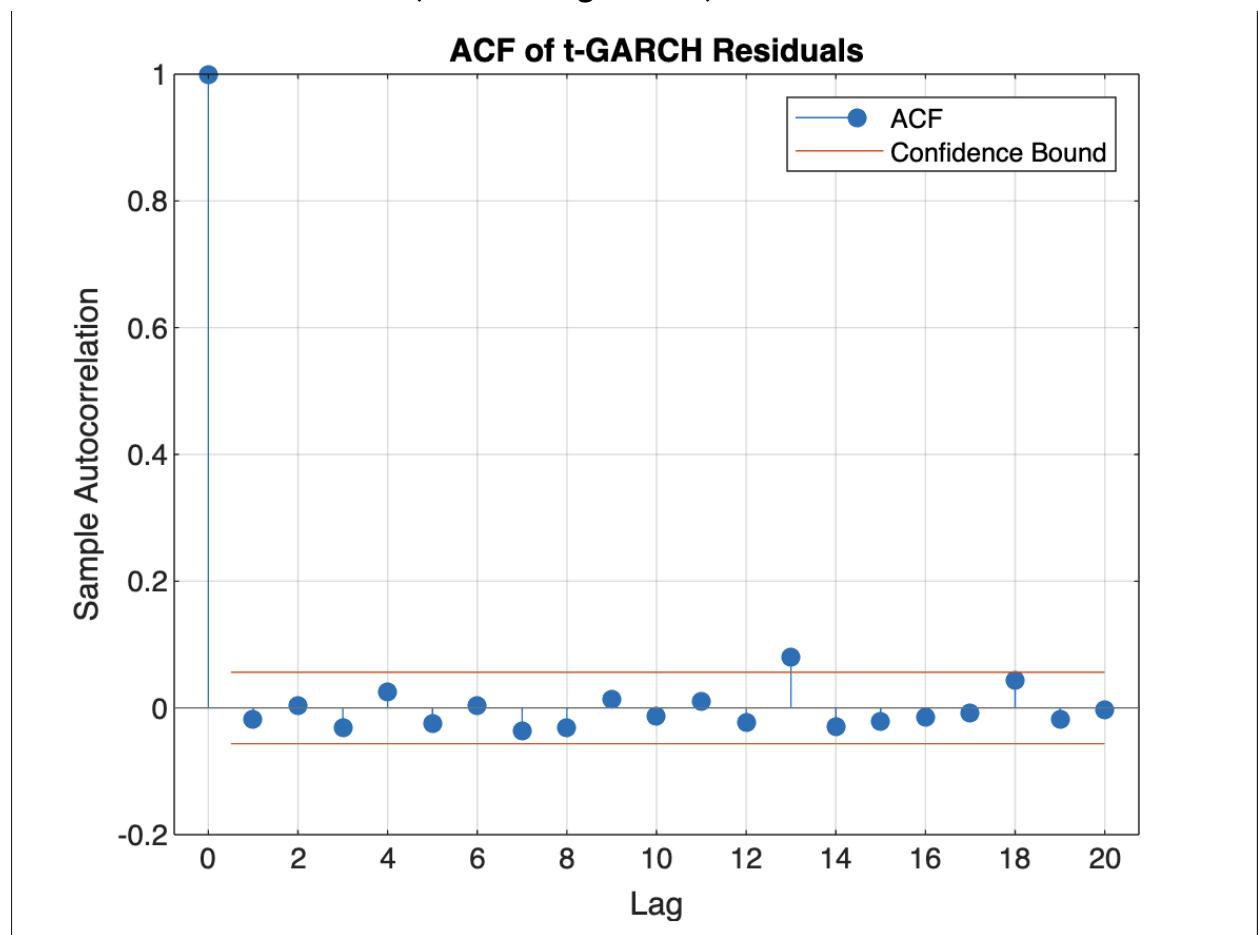
ACF of t-GARCH Residuals:



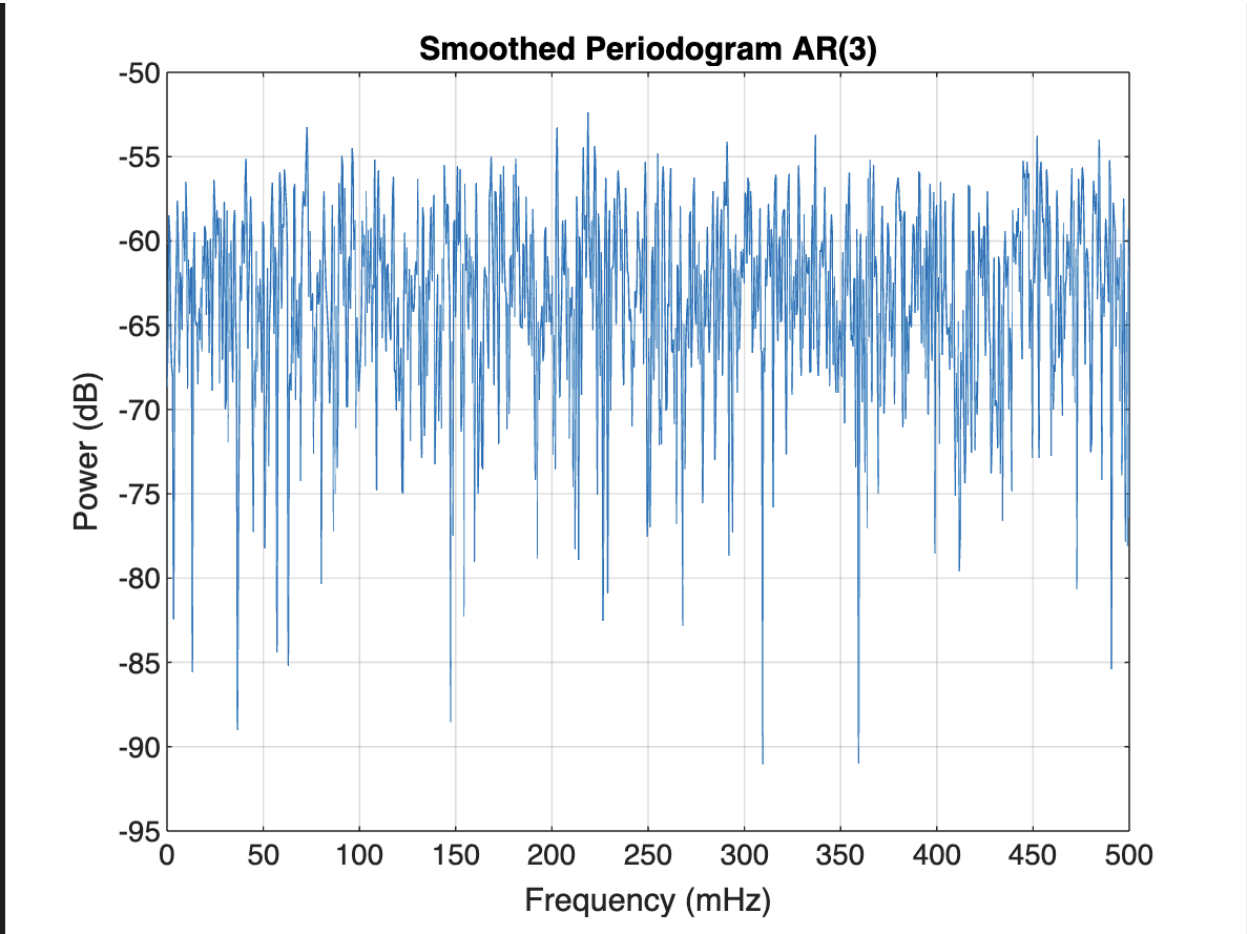
Normal Q-Q Plot of T-GARCH Residual (based of logReturns):



ACF of t-GARCH Residuals (based of logReturns):



Smoothed Periodogram:



AIC Table for Model Comparison:

Model	AIC
{'ARMA(1,1)'}	-5957.8
{'ARMA(2,1)'}	-5956
{'ARMA(2,2)'}	-5954.6

Model Fitting (ARMA(2,1)):

	Value	StandardError	TStatistic	PValue
Constant	-0.00060163	0.0010421	-0.57734	0.56371
AR{1}	-0.65089	0.46805	-1.3906	0.16434
AR{2}	0.012011	0.031444	0.38198	0.70247
MA{1}	0.63415	0.46969	1.3502	0.17697

Variance	0.00050464	1.1274e-05	44.761	0
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Inverse Roots ARMA(2,1) - Didn't work

Inverse Roots ARMA(2,2):

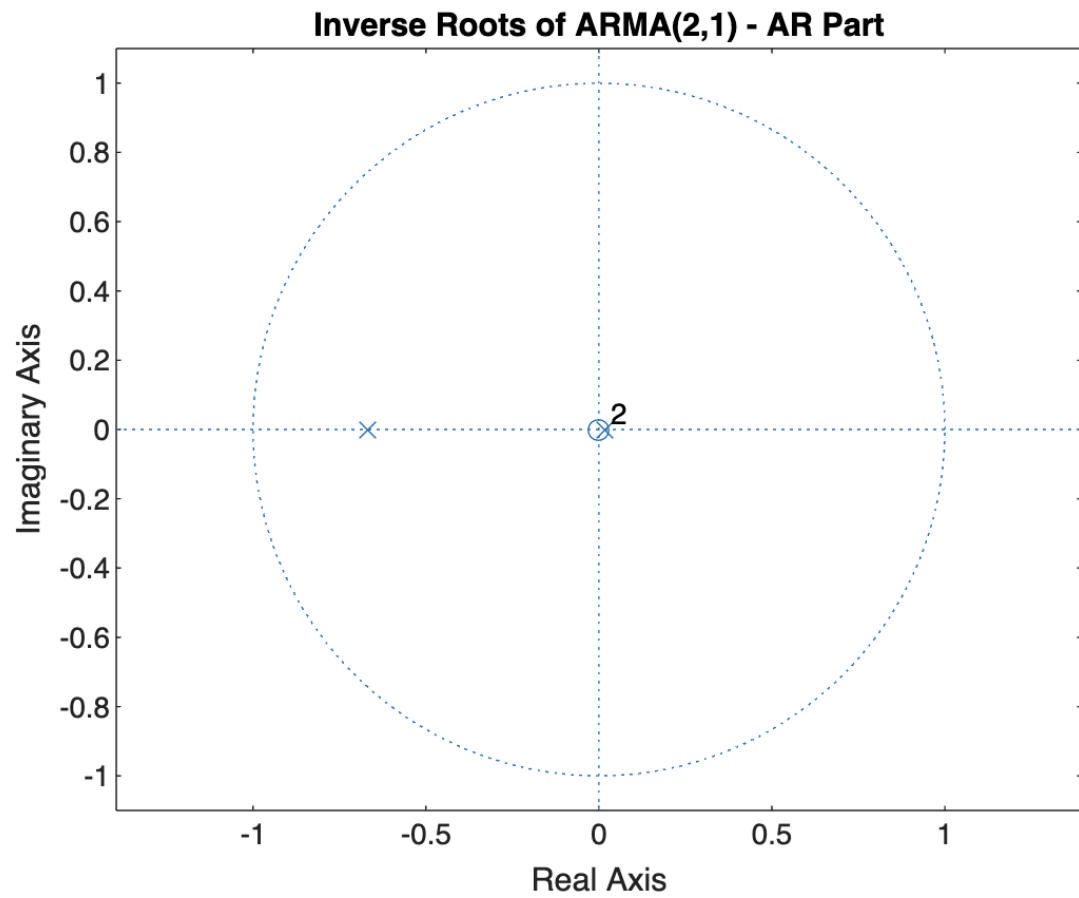
ARIMA(2,0,2) Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.00012372	0.00030499	-0.40565	0.685
AR{1}	0.063939	0.60663	0.1054	0.91606
AR{2}	0.61128	0.36378	1.6804	0.092883
MA{1}	-0.096057	0.61115	-0.15717	0.87511
MA{2}	-0.60483	0.37159	-1.6277	0.10359
Variance	0.00050441	1.123e-05	44.915	0

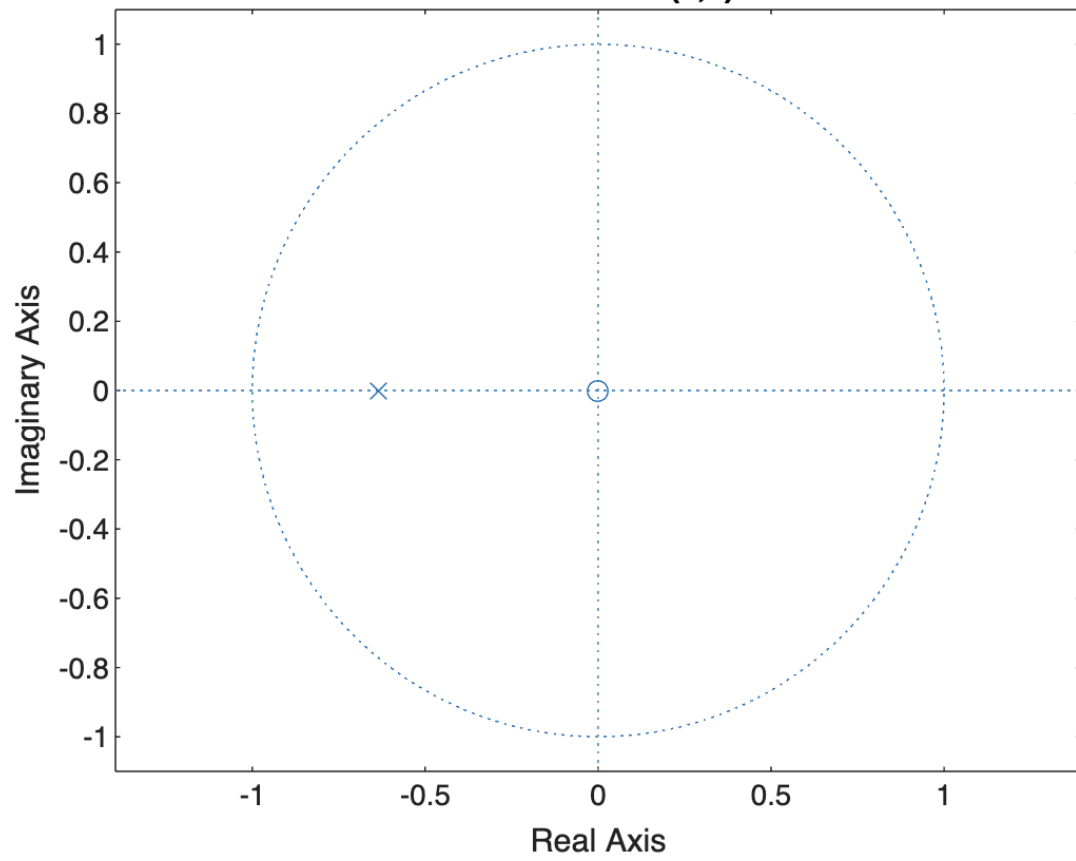
Likelihood Ratio Test between ARMA(2,2) and ARMA(2,1)L

Likelihood Ratio Test p-value: 0.4492

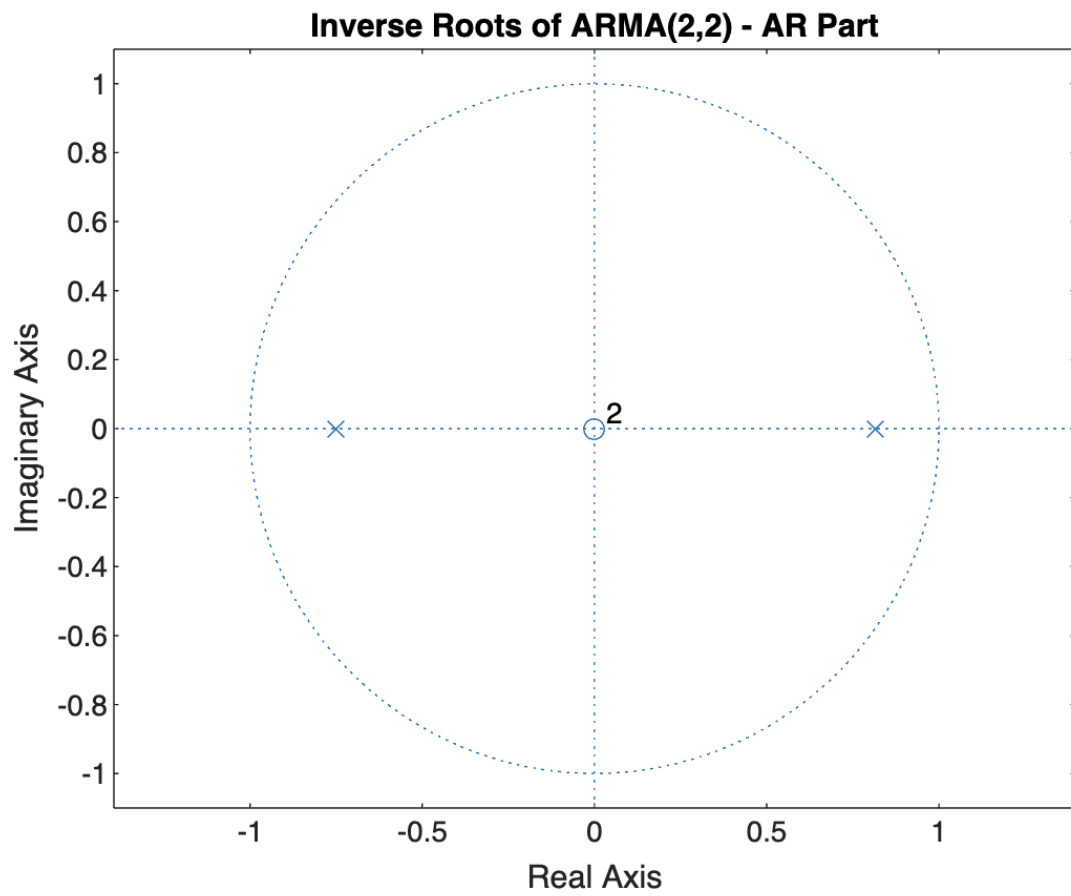
Inverse Roots of ARMA(2,1) - AR Part and MA Part:

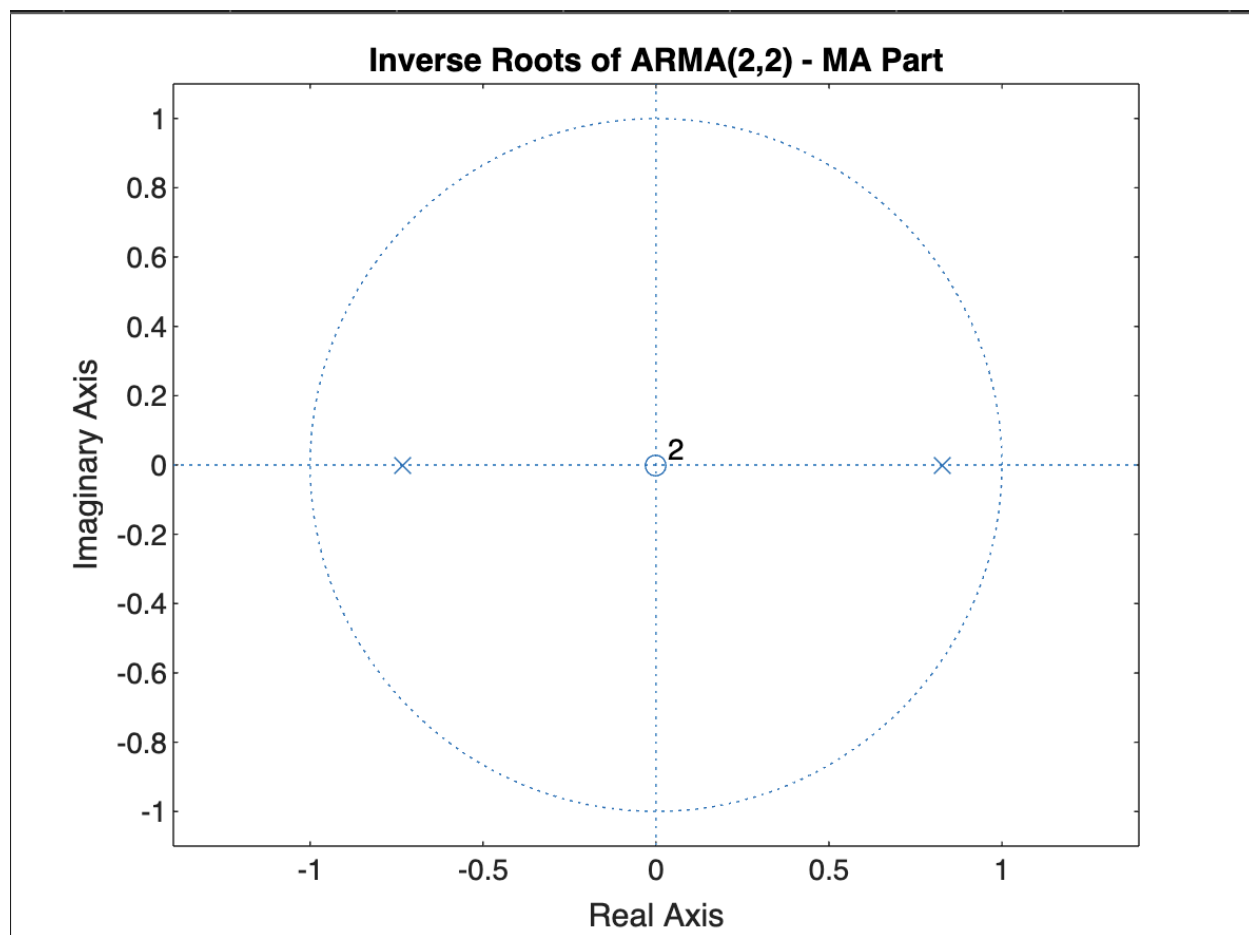


Inverse Roots of ARMA(2,1) - MA Part



Inverse Roots of ARMA(2,2) - AR and MA Part:





EGARCH(1,1) Conditional Variance Model (Gaussian Distribution):

	Value	StandardError	TStatistic	PValue
Constant	-0.032444	0.017832	-1.8194	0.068844
GARCH{1}	0.99538	0.0023858	417.21	0
ARCH{1}	0.061922	0.0075463	8.2056	2.2941e-16
Leverage{1}	-0.027469	0.0095519	-2.8758	0.0040305

Wavelet-Based Volatility Analysis:

